DMV Practical 3: Real Estate Data Preprocessing

Source: Your CL-1 practical journal

Aim: To preprocess a real estate dataset by cleaning missing values, merging datasets, filtering records, and preparing the data for further analysis.

In this practical, we work with property information such as property type, neighborhood, number of bedrooms, bathrooms, area, year built, and sale price.

We'll study it cell by cell, in simple paragraphs, focusing on what the cell does, why it is used, and what the output means—without explaining every code line separately.

Cell 1: Importing Libraries

This cell prepares the notebook by importing the Python libraries required for data preprocessing. Libraries such as Pandas and NumPy are commonly used to load, manage, and process the real estate dataset.

Cell 2: Load the Dataset

Purpose: This cell loads the real estate CSV file into a Pandas DataFrame named df.

The head() function displays the first five records of the dataset so we can understand its structure. The shape function displays the total number of rows and columns.

Output: The dataset contains 100 rows and 9 columns. It includes details such as Property ID, Property Type, Neighborhood, Bedrooms, Bathrooms, Area, Year Built, Sale Price and Sale Year.

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Cell 3: Clean Column Names and Handle Missing Values

This cell cleans the column names and handles missing values in the dataset. Column names are converted into lowercase, extra spaces are removed, and spaces or special characters are replaced or removed to make the names easier to use in Python.

It also checks for missing values. The dataset initially contains 2 missing values in the neighborhood column and 3 in the bathrooms column. These are handled by filling numerical missing values in bathrooms with the median and categorical missing values in neighborhood with the mode. Finally, it displays the missing-value count again, which should be zero for all columns.

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Remember for viva:

- head() – Displays the first five records.
- shape – Returns rows and columns.
- median – Middle value of numerical data.
- mode – Most frequently occurring value.
- fillna() – Fills missing values.

Cell 4: Data Merging

Purpose: This cell combines two datasets to add more information about each property's neighborhood.

Here, a new DataFrame called neighborhood_data is created, containing neighborhood names and their average income. The pd.merge() function combines this information with the original real estate dataset using the common column neighborhood. The how="left" option ensures that all records from the original real estate dataset are retained, even if a matching neighborhood is not found in the second dataset.

Finally, display(df.head()) shows the first five records after merging. The resulting dataset includes an additional column called avg_income.

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Remember for viva:

- Data merging: Combining two datasets using a common column.
- pd.merge(): Used to merge DataFrames.
- on="neighborhood": Specifies the common column.
- how="left": Keeps all records from the left (original) dataset.
- avg_income: Newly added column.

In simple words: Cell 4 adds neighborhood average income information to the original property dataset.

Cell 5: Filter and Subset Data

Purpose: This cell is used to filter the real estate dataset and select specific property records and important columns.

Explanation:

In this cell, we perform three operations on the real estate dataset.

1. Filter properties by sale year: The dataset is filtered to display only those properties that were sold between 2021 and 2023. This helps us analyze properties sold during a particular time period.

2. Select only houses: The dataset is filtered to display records where the property type is House. This helps us focus only on house properties and exclude other property types.

3. Select important columns: A subset of the dataset is created by selecting only important columns, such as property type, neighborhood, bedrooms, area, and sale price. This makes the dataset smaller and easier to analyze.

In simple words: This cell helps us extract specific property records based on sale year and property type, and select only the columns required for further analysis.

Important Viva Questions

1. What is data filtering?
Data filtering means selecting specific rows from a dataset based on given conditions.

2. What is subsetting?
Subsetting means selecting specific rows or columns from a dataset.

3. Which operator is used to combine conditions?
The & operator is used to combine multiple conditions.

4. Why do we filter data?
To focus on relevant records and make analysis easier.

Cell 6: One-Hot Encoding

Purpose: This cell converts categorical data into numerical data so that it can be used for machine learning algorithms.

Explanation:

In this cell, we use One-Hot Encoding to convert the categorical columns property_type and neighborhood into separate numerical columns.

For example, the property_type column may contain values such as House, Apartment, and Villa. One-Hot Encoding creates separate columns for each category, such as property_type_House, property_type_Apartment, and property_type_Villa.

Similarly, the neighborhood column is converted into separate columns for each neighborhood. Each new column contains 1 or 0, where 1 represents that the category is present and 0 represents that it is absent.

The encoded dataset is stored in df_encoded. Finally, the cell displays the first five rows and prints all the newly created column names.

In simple words: One-Hot Encoding converts text-based categories into numerical 0s and 1s, making the dataset suitable for machine learning.

Important Viva Questions

1. What is One-Hot Encoding?
It is a technique used to convert categorical data into numerical values (0 and 1).

2. Why is One-Hot Encoding used?
To make categorical data understandable to machine learning algorithms.

3. Which function is used?
pd.get_dummies().

4. What does 1 represent?
The category is present.

5. What does 0 represent?
The category is absent.

6. What is the use of dtype=int?
It ensures that the encoded values are stored as integers.

Remember: One-Hot Encoding creates separate columns for categories instead of assigning them numbers like 1, 2, and 3.

Cell 7: Data Analysis and Feature Creation

Purpose: This cell calculates the average sale price of properties in different neighborhoods and property types. It also calculates the price per square foot.

Explanation:

In this cell, we perform three main operations.

1. Average Sale Price by Neighborhood: The dataset is grouped according to neighborhood, and the average sale price of properties in each neighborhood is calculated. The results are arranged in descending order, so the neighborhood with the highest average sale price appears first.

2. Average Sale Price by Property Type: The dataset is grouped according to property type, such as House, Apartment, or Villa. The average sale price of each property type is calculated and sorted from highest to lowest.

3. Average Price per Square Foot: A new column called price_per_sq_ft is created. It calculates the price of a property for each square foot by dividing its sale price by its total area. Finally, the first five property IDs and their price per square foot are displayed.

In simple words: This cell helps us understand which neighborhoods and property types have higher average prices and how much a property costs per square foot.

Important Viva Questions

1. What is groupby()?
It is used to group data based on a particular column.

2. What is mean()?
It calculates the average value.

3. What is sort_values()?
It arranges values in ascending or descending order.

4. What does ascending=False mean?
It sorts the values from highest to lowest.

5. What is price per square foot?
It is the sale price of a property divided by its area in square feet.

6. Why do we create a new column?
To add a useful feature that helps in further data analysis.

Important formula

Remember: This cell performs data analysis and feature creation on the real estate dataset.

Cell 8: Outlier Detection Using IQR Method

Purpose: This cell detects outliers in the real estate dataset, especially in sale price and property area.

Explanation:

In this cell, we use the IQR (Interquartile Range) method to identify unusually high or low values in the dataset.

First, a function named detect_outliers() is created. It calculates Q1 (first quartile) and Q3 (third quartile) to find the IQR. Using these values, it calculates the lower and upper limits. Any value below the lower limit or above the upper limit is considered an outlier.

Next, this function is applied to the sale_price_ column to detect properties with unusually high or low sale prices. It displays the property IDs, sale prices, and calculated limits.

The same function is then applied to the area_sq_ft column to identify properties with unusually small or large areas.

In simple words: This cell finds unusual values in property prices and property areas that are very different from the other values in the dataset.

Important Formula

$$IQR = Q3 - Q1$$

$$Lower\ Bound = Q1 - 1.5 \times IQR$$

$$Upper\ Bound = Q3 + 1.5 \times IQR$$

Important Viva Questions

1. What is an outlier?
An outlier is a value that is significantly different from other values in a dataset.

2. What is IQR?
IQR stands for Interquartile Range. It is the difference between Q3 and Q1.

3. What are Q1 and Q3?
Q1 is the 25th percentile, and Q3 is the 75th percentile.

4. Why do we detect outliers?
To identify unusual values that may affect data analysis and machine learning results.

5. Does this cell remove outliers?
No. It only detects and displays outliers. It does not remove or modify them.

Remember: IQR is a statistical method used to detect unusual values in a dataset.

Cell 9: Remove Outliers

Purpose: This cell removes unusual values (outliers) from the real estate dataset to make the data cleaner.

Explanation:

In this cell, we create a new dataset called df_clean by keeping only those property records whose sale prices and areas fall within the lower and upper limits calculated in the previous cell.

Properties with sale prices outside the allowed range or areas outside their allowed range are removed. After removing these outliers, the cell displays the dataset shape before and after cleaning.

The shape tells us the number of rows and columns remaining in the dataset. By comparing both shapes, we can understand how many records were removed.

In simple words: This cell removes properties with unusually high or low sale prices and areas from the dataset.

Important Viva Questions

1. Why do we remove outliers?
To reduce the effect of unusual values on data analysis.

2. Which method is used to detect outliers?
The IQR method.

3. What is df_clean?
It is the new dataset after removing outliers.

4. What does shape display?
It displays the number of rows and columns.

Cell 10: Display Final Cleaned Dataset

Purpose: This cell displays the final cleaned dataset and checks its information.

Explanation:

In this cell, we display the first five records of the cleaned dataset using head(). This helps us check whether the data looks correct after preprocessing and outlier removal.

Next, df_clean.info() displays important information about the final dataset, such as the total number of entries, column names, data types, and non-null values. It helps us verify the structure of the cleaned dataset.

In simple words: This cell checks and displays the final dataset after all the preprocessing steps.

Important Viva Questions

1. What is head()?
It displays the first five rows of a dataset by default.

2. What is info()?
It displays dataset information, including columns, data types, and non-null values.

3. What is the purpose of the final cleaned dataset?
It is prepared for further data analysis and machine learning.

Remember: Cell 9 removes outliers, while Cell 10 displays and verifies the final cleaned dataset.

Practical 4: Data Visualization using Matplotlib

Title: Analyzing Air Quality Index (AQI) Trends in a City.

Cell 1: Import Libraries

In this cell, we import two important libraries. Pandas is used to handle and analyze the AQI dataset, while Matplotlib is used to create graphs and visualize the data.

The figure size setting defines the graph size as 10 inches wide and 5 inches high. The grid setting displays grid lines in graphs, making values easier to read.

In simple words: This cell imports the libraries required for data analysis and graph creation and sets the default appearance of graphs.

Important Viva Questions

1. What is Pandas?
   A Python library used for data manipulation and analysis.
2. What is Matplotlib?
   A Python library used for creating graphs and charts.
3. What is the use of plt.rcParams?
   It is used to set default graph properties.
4. What is AQI?
   AQI stands for Air Quality Index. It indicates the level of air pollutionIn this cell, we load the air quality dataset named City_Air_Quality.csv into a Pandas DataFrame called df.
5. The try-except block is used for error handling. If the CSV file is available, it loads successfully and displays a confirmation message. If the file is missing, it displays an error message instead.
6. Finally, df.head() displays the first five records of the dataset so we can check the data.

   (The first five records contain these columns 😊

   CELL3

   This cell performs an initial examination of the dataset. It checks the total number of rows and columns, displays all column names, and identifies the data types of each column.

   It also checks whether any values are missing and generates summary statistics such as count, mean, standard deviation, minimum and maximum values. This helps us understand the dataset before creating visualizations.

   Important Viva Questions

1. What is df.shape?
   It returns the number of rows and columns.
2. What is df.dtypes?
   It displays the data type of each column.
3. What is isna()?
   It identifies missing or null values.
4. What is describe()?
   It generates summary statistics of the dataset.
5. What does T mean?
   It transposes the output, switching rows and columns.

   In simple words: This cell checks the dataset's structure, data types, missing values and statistical information.

   Cell 4: Identify Date, AQI and Pollutant Columns

Cell 4: Explanation

This cell identifies three important types of columns in the AQI dataset: the date column, AQI column and pollutant columns.

The program searches the column names and automatically identifies the date and AQI columns. It then converts the date column into a proper date format using pd.to_datetime(). After that, it searches for pollutant columns such as PM2.5, PM10, CO, NO2, SO2 and O3.

This makes the dataset ready for further analysis and visualization of air quality trends.

Important Viva Questions

1. What is date_candidates?
   It stores the possible date-related column names.

2. What is aqi_candidates?
   It stores the column names containing AQI.

3. What is pd.to_datetime()?
   It converts date values into datetime format.

4. What is pollutant_keywords?
   It contains common names of air pollutants.

5. Name three pollutants used in this practical.
   PM2.5, PM10 and CO.

6. Why do we identify these columns?
   To analyze AQI and pollutant trends over time

7. **Cell 5: AQI Trend Over Time**
8. **Aim**
9. To visualize how the Air Quality Index (AQI) changes over different dates.
10. **Explanation**
11. This cell creates a line graph to show the changes in AQI over time. First, it removes records where the date or AQI value is missing and arranges the remaining records in date order.
12. The graph displays dates on the X-axis and AQI values on the Y-axis. Circular markers represent individual AQI readings, while the connecting line helps us understand whether air quality is improving, worsening or fluctuating over time.
13. The date labels are rotated by 45 degrees to make them easier to read. Finally, the graph is displayed using Matplotlib.
14. **Cell 6: Pollutant Trends Over Time**
15. **Aim**
16. To visualize how different air pollutants change over time using line graphs.
17. **Explanation**
18. This cell creates separate line graphs for each pollutant present in the dataset.
19. First, it removes rows where the date is missing and sorts the data according to date. Then, it uses a loop to check each pollutant column. If the column contains numeric values, it creates a line graph for that pollutant.
20. The graph displays the date on the X-axis and the pollutant value on the Y-axis. Each graph has a title showing the pollutant name. The legend identifies the pollutant, and the date labels are rotated for better readability.

Graph 1: PM2.5 Trend Over Time

Shows how fine particulate matter (PM2.5) changes over the dates.

Graph 2: PM10 Trend Over Time

Shows how PM10 particle concentration changes over the dates.

Graph 3: CO Trend Over Time

Shows how carbon monoxide concentration changes over the dates.

The three graphs show the changes in different air pollutants over time. The first graph (PM2.5 Trend Over Time) shows how the concentration of fine particles (PM2.5) changes on different dates. The second graph (PM10 Trend Over Time) represents the changes in PM10 particle concentration over time. The third graph (CO Trend Over Time) shows how carbon monoxide levels vary across the dates. All three graphs use dates on the X-axis and pollutant values on the Y-axis, helping us understand the variations in air pollution over time.

Cell 7: Average AQI by Month

Aim

To calculate and visualize the average AQI for each month using a bar graph.

Explanation

This cell first removes rows with missing date or AQI values and sorts the data by date. It then extracts the month and year from each date and groups the AQI values month-wise. After grouping, it calculates the average AQI for each month using the mean() function. Finally, it displays the results in a bar graph, where the X-axis represents the months and the Y-axis represents the average AQI.

Bar graph.

2. What is the purpose of groupby()?
It groups data based on a particular column.

3. What does mean() calculate?
It calculates the average value.

4. What is the use of to_period("M")?
It extracts the month and year from the date.

5. What does the X-axis represent?
Month.

6. What does the Y-axis represent?
Average AQI.

Cell 8: AQI Distribution – Boxplot and Histogram

Aim

To understand the distribution and frequency of AQI values using a boxplot and histogram.

Explanation

This cell creates two graphs to analyze the distribution of AQI values. First, it removes missing AQI values using dropna() and displays a boxplot, which helps identify the median, spread and possible outliers in AQI data. Next, it creates a histogram by dividing AQI values into 15 groups (bins) and showing how frequently the values occur. Both graphs help us understand the overall distribution of air quality in the city.

Graph 1: AQI Distribution (Boxplot)

This graph shows the spread of AQI values, including the median, lower and upper quartiles, and possible outliers. It helps us understand how much AQI values vary in the dataset.

Graph 2: AQI Histogram

This graph shows how frequently different AQI values occur. The X-axis represents AQI values, and the Y-axis represents their frequency. Taller bars indicate that more readings fall within that AQI range

4. What is the use of bins=15?
It divides the AQI values into 15 intervals.

5. What does frequency mean?
The number of times values occur in a particular range.

6. What is an outlier?
A value that is unusually different from other values.

7. What is the use of dropna()?
It removes missing AQI values.

In simple words: The boxplot shows the spread and possible outliers of AQI values, while the histogram shows how frequently different AQI values occur.

Cell 9: AQI vs Pollutants – Scatter Plots and Correlation

Aim

To study the relationship between AQI and different air pollutants using scatter plots.

Explanation of the 3 Graphs

This cell creates three scatter plots to understand the relationship between AQI and different pollutants.

GRAPH1-This graph shows the relationship between PM2.5 and AQI. The points show a strong positive relationship, meaning that when PM2.5 increases, AQI generally also increases.

GRAPH2This graph shows the relationship between PM10 and AQI. It indicates a weak positive relationship, meaning that an increase in PM10 is only slightly associated with an increase in AQI.

Correlation: 0.117

GRAPH3-This graph shows the relationship between carbon monoxide (CO) and AQI. It indicates a weak positive relationship, meaning that CO and AQI have a slight positive association in this dataset.

Correlation: 0.235

Correlation measures the strength and direction of the relationship between two variables.

- +1: Perfect positive correlation.
- 0: No linear correlation.
- -1: Perfect negative correlation.

Important Viva Questions

1. Which graph is used in this cell?
Scatter plot.

2. What is a scatter plot?
A graph that shows the relationship between two numerical variables.

3. What is correlation?
It measures the strength and direction of the relationship between two variables.

4. Which pollutant has the highest correlation with AQI?
PM2.5, with a correlation of 0.958.

5. What is the use of .corr()?
It calculates the correlation between two numerical columns.

6. What is the use of alpha=0.7?
It controls the transparency of the plotted points.

7. Does correlation prove causation?
No. Correlation does not prove that one variable causes another.